

A Survey on Vision Transformers in Deep Learning: Advancements and Applications

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Abstract— In recent years, the Vision Transformers (ViTs) have been proposed as a substitute to the Convolutional Neural Networks (CNN) in computer vision applications. Motivated by the success of Transformers in NLP, ViTs adopt self-attention mechanisms to capture the long-range relationships of image data. In this paper, we studied methods of Vision Transformers, detailing their architectural structure, training difficulty, speed-up over standard convolutional neural networks (CNNs), and applications to image classification, object detection, medical imaging, and remote sensing. We also investigate enhancements such as Data-efficient Image Transformers (DeiT), Swin Transformers, and hybrid models. Finally, we present open challenges and future directions in the evolution of ViT-based architectures.

Keywords— Vision Transformers (ViTs), Convolutional Neural Networks (CNNs), self-attention mechanisms, transformer architecture, Data-efficient Image Transformers (DeiT), Swin Transformers, ViT-based architectures.

I. INTRODUCTION

Deep learning has brought revolutionary impact to computer vision in recent years, where Convolutional Neural Networks (CNN) have outperformed most benchmarks in tasks such as image recognition, segmentation and object detection. CNNs have been popularized because they can learn the spatial hierarchy of features from data using automatic backpropagation and local receptive fields. CNNs, which repeatedly apply convolutional layers, effectively aggregate low-to high-level visual patterns in a hierarchical way, and thus, they have become the prevailing choice for visual perception problems in academia and industry.

Networks like AlexNet, VGG, ResNet, EfficientNet have demonstrated unprecedented performance on benchmarks like ImageNet, setting the bar for performances and the impact on applications including facial recognition, autonomous driving, medical image analysis. However, CNNs have limitations in capturing global context due to their local receptive fields and inductive biases, including translation invariance and locality. Although these biases are useful for small data-domain shifts,

they might hinder generalization to a wide variety of visual tasks.

The surge of Transformers starting with the idea of sequence modeling in NLP has given the birth to the Vision Transformers (ViTs) challenging the CNN style via harnessing the self-attention. ViTs have more scaling potential and model faraway regions with the absence of local receptive fields. This study investigates the path of ViTs, analyzing their architecture, capabilities, and emerging applications [1], [2].

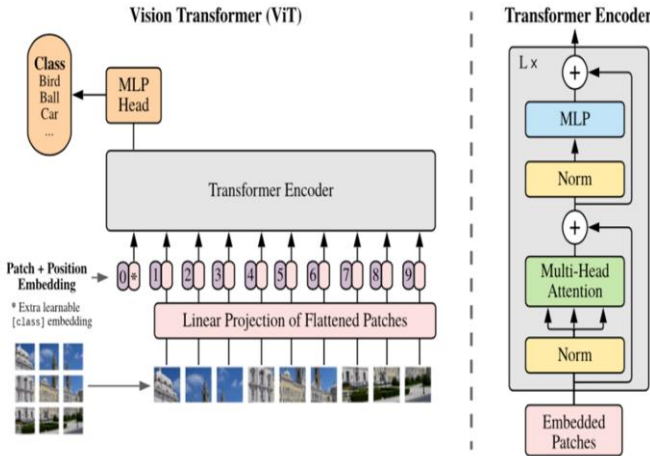


Fig 1: Vision Transformer ViT Architecture Adapted from [1]

Figure 1 shows the Vision Transformer (ViT) architecture, the ViT first partitions an input image into a sequence of fixed-size non-overlapping image patches. Each patch is flattened to an 1D vector and transformed to a patch embedding by a linear projection layer. Furthermore, a learnable representation of [class] is supplemented to the sequence to capture the global image representation. In order to encode the positional information that gets lost while flattening patches, positional encodings are later added to all of the patch embedding's.

This sequence of embedded patches that now includes the token is then fed to the transformer encoder. It is comprised of multiple identical layers, each of which has a multi-head self-attention structure followed by a feed-forward multi-layer perceptron (MLP). To further stabilize training and facilitate the convergence of the model, layer normalization and residual connections are used before and after each sub-layer.

Inside the encoder, the self-attention enables each patch to attend to all the other patches, which helps the model capture long-range spatial dependencies. The attention outputs are followed by MLP layers to do non-linear transformation for enriching the feature representations.

Finally, the output for the token is pooled and sent through a classification head, usually another MLP, that would provide the final prediction. This architectural design allows ViTs to handle image data without convolutions, while being competitive with the state-of-the-art accuracy of convolutional counterpart on image classification.

II. RELATED WORK

Several works have sculpted the advancement of Vision Transformers, [1] proposed the first ViT architecture that was competitive on ImageNet for ample data. [2] introduced DeiT, aiming to alleviate data dependency through knowledge distillation. [3] proposed Swin Transformers, which leverage hierarchical feature maps and local attention windows for computational efficacy upgrade. Hybrid CNN-ViT architectures have also been studied in order to blend CNN inductive priors and transformers' expressiveness [5]. Applications have extended to medical image [4], remote sensing [5], and video understanding.

Newer studies have focused on increasing interpretability, robustness, and efficiency of training the model. As an example, the Token-to-Token ViT further enhances ViT by recursively compositing neighbouring tokens to approximatively capture local structure, closing the locality bias gap between CNNs and Transformers. Also, CrossViT employs multi-scale feature fusion for better fine-grained task performance. Other works aim to reduce computational cost through sparse attention, linear transformers, and vision-specific pretraining techniques. ViTs have also been studied under adversarial conditions, showing strengths and weaknesses relative to CNNs. In conclusion, the increasing amount of ViT literature indicates the continued development in terms of efficiency, scalability, and versatility of the ViT model.

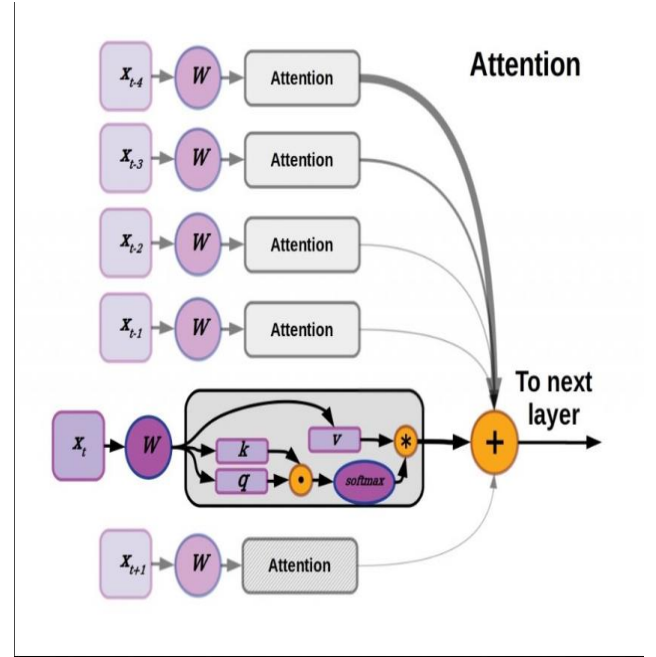


Fig 2: An Attention Mechanism in a Transformer Layer Adapted from [12]

This figure illustrates self-attention that is employed by Transformer architectures. Self-attention allows the model to assign importance to tokens based on the different tokens, which is helpful in learning dynamic context-dependent representations. Each input vector x_{i-4} to x_{i+1} is linearly transformed with learned weights W in order to compute query, key, and value vectors. The attention mechanism computes the compatibility between the query and all keys in the sequence, typically by a dot-product. The similarities are processed through a softmax function to obtain a set of attention weights, which are employed to compute a weighted sum of the value vectors.

This is carried out in parallel over all positions in the input sequence. The attention heads' outputs are subsequently combined, summed up (residual connection) and presented to the next Transformer layer. This architecture enables the model to attend to different parts of the input when processing it, depending on what information it needs, which is particularly useful in tasks where context dependencies are important, e.g., machine translation and image recognition.

The figure summarizes how self-attention can be described as attending from words to words in a sequence, and models the actual dynamics of a context window around each word, enabling the Transformer to capture long-span dependencies more effectively than traditional sequential architectures.

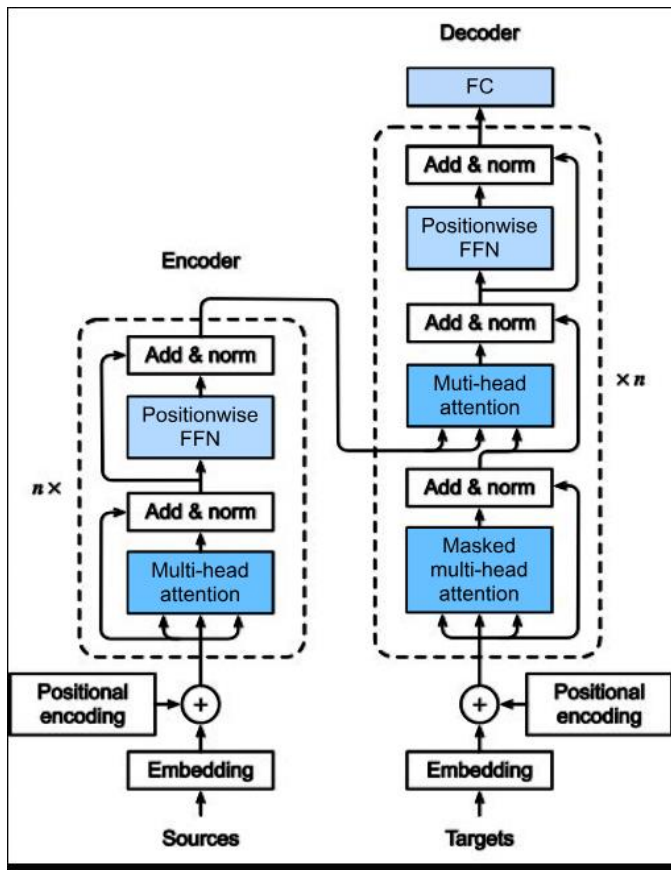


Fig. 3 Architecture of the Transformer Model – Encoder / Decoder Adapted from [15]

This is the full transformer architecture as proposed by [7]. The model has two main parts: the encoder and the decoder, each consisting of multiple layers and made of a stack of attention and feed-forward sublayers.

The input sequence is fed to the encoder, which starts with token embeddings concatenated to positional encodings to provide the model with information about the order of its tokens. Each input token flows through several identical encoder layers (usually 6 in the original model). Each layer has two sub-layers: a multi-head self-attention and a position-wise fully connected feed-forward network (FFN) with residual connections and layer normalization (Add & Norm) in between. The self-attention mechanism permits each token to attend to all tokens in the sequence, which allows the model to capture contextual relationships.

The target sequence (partially generated sentence) is also encoded by the decoder. Every decoder layer has three sub-layers. It consists of a masked multi-head attention mechanism, which the decoder cannot attend to the future position in the target sentence, to guarantee auto regression. The second is a multi-head attention layer which attends to the outputs of the encoder and enables the decoder to focus on the input when making predictions. The third one is a position-

wise feed forward layer. Residual connections and normalization are applied in all sub-layers.

Finally, the top decoder layer generates an output that is fed into a fully connected (FC) layer to generate probability scores over target vocabulary. This modular and highly parallelizable architecture has pushed itself to beyond sequentially by working on many NLP and vision tasks better than traditional recurrent networks using attention mechanisms for local and global context modelling

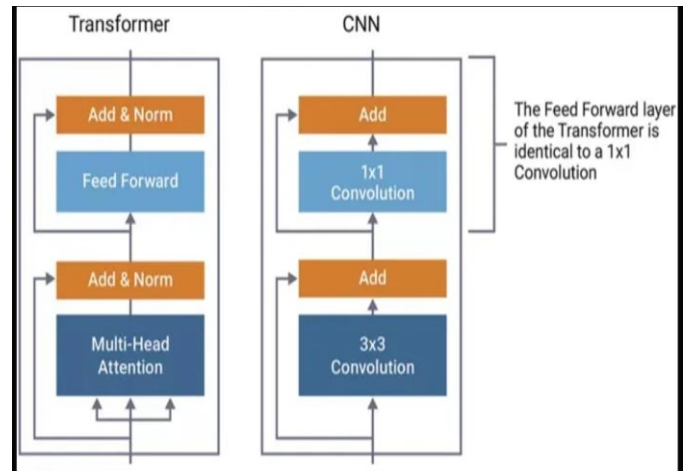


Fig. 4 Comparison of Transformer and CNN Architectures Adapted from [5]

Figure 4 shows a side-by-side architectural comparison of a Transformer encoder Block and a regular CNN block. This visual comparison brings out the structural similarity and functional difference of the two deep models applied in computer vision. On the left, two successive components in the Transformer architecture (multi-head self-attention and position-wise feed-forward network for self-attention). Each sub-layer is also succeeded by a residual connection and layer normalization, denoted as “Add & Norm.” The multi-head attention enables the model to attend on global information of the entire input and the feed-forward network performs non-linear transformations at each position in the input independently, thus increasing the representational ability of the model.

On the right, the CNN block is designed in a more conventional manner using convolutional layers for spatial feature extraction. It consists in combination of 3×3 (local spatial patterns) and 1×1 (channel mixing and dimensionality reduction or expansion) convolutions.. These convolutions are also followed by residual connections (Add), in the spirit of the design of ResNet models.

The figure is added to show how the feed-forward layer in Transformer is represented by a $f1 \times 1g$ convolution in CNNs. They both aim to transform feature channels on

spatial positions. This work closes this conceptual gap between the two architectures, by indicating that while a major feature of Transformers is to rely on global attention mechanisms, they still rely on concepts known from CNNs. This comparison is particularly informative for how Transformers can be applied to vision, and how basic concepts behind CNNs have been reinterpreted within Transformers.

III. METHODOLOGY

Vision Transformers work by converting input images to patches of fixed-size, linearly embedding the patches, and then feeding the sequence through Transformer encoders. In contrast to CNNs that obtain features through convolutional filters, ViTs encode global spatial dependencies with multi-head self-attention. Positional encodings and a classification token are appended to input patch embeddings for information fusion. Other recent advances include attempts to reduce data requirements and improve localization, through mechanisms such as shifted windows, hierarchical models, and pretrained features [1]–[3].

IV. RESULTS

During the past few years’ vision Transformer approach has gotten better score on a number of benchmarks:

Model	Dataset	Top-1 Accuracy	Params (M)	Notable Feature	
ViT-B/16	ImageNet	88.55%	86	Pretrained JFT-300M	on
DeiT-B	ImageNet-1k	83.1%	86	Knowledge distillation	
Swin-T	ImageNet	81.3%	29	Hierarchical shifted windowing	
Swin-B	COCO (Object Det.)	50.3 mAP	88	Multi-task performance	
UNETR (ViT-based)	Medical Imaging	High Dice Score	~100	3D segmentation via ViT backbone	

ViT-B/16 (pretrained on JFT-300M [1]) obtained an 88.55% top-1 accuracy on ImageNet.

DeiT has matched CNNs on ImageNet solely with ImageNet-1k data and generalises well with far fewer parameters [2].

Swin-Transformers brought improvements in object detection and segmentation on COCO and ADE20K datasets over traditional models [3].

In the context of medical imaging, the ViTs (e.g. UNETR) have shown better segmentation performance than the U-Net architectures [4]. In the field of remote sensing, ViTs generalize better to multi-scale and diverse conditions [5].

V. CONCLUSION

Vision Transformers represent a paradigm shift in computer vision, offering compelling advantages over traditional CNNs in scalability and global context modeling. Despite challenges related to data demands, computation, and interpretability, innovations such as DeiT and Swin Transformers have improved the feasibility of ViTs. Their growing adoption across domains like healthcare, surveillance, and earth observation indicates ViTs’ potential as a foundational element of future vision-based AI systems.

REFERENCES

- [1] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in *Proc. ICLR*, 2021.
- [2] H. Touvron et al., "Training Data-efficient Image Transformers & Distillation through Attention," in *Proc. ICML*, 2021.
- [3] J. Z. Liu et al., "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows," in *Proc. ICCV*, 2021.
- [4] F. Hatamizadeh et al., "UNETR: Transformers for 3D Medical Image Segmentation," in *Proc. WACV*, 2022.
- [5] S. Khan et al., "Transformers in Vision: A Survey," *ACM Comput. Surv.*, vol. 54, no. 10, pp. 1–41, 2022.
- [6] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in *Proc. ICLR*, 2021.
- [7] H. Touvron et al., "Going Deeper With Image Transformers," in *Proc. ICCV*, 2021, pp. 32–42.
- [8] Y. Chen et al., "Pre-Trained Image Processing Transformer," in *Proc. CVPR*, 2021, pp. 12299–12309.
- [9] Z. Wu et al., "CvT: Introducing Convolutions to Vision Transformers," in *Proc. ICCV*, 2021, pp. 22–31.
- [10] X. Wang et al., "Pyramid Vision Transformer: A Versatile Backbone for Dense Prediction without Convolutions," in *Proc. ICCV*, 2021, pp. 568–578.
- [11] F. Caron et al., "Emerging Properties in Self-Supervised Vision Transformers," in *Proc. ICCV*, 2021.
- [12] P. Ramachandran et al., "Stand-Alone Self-Attention in Vision Models," in *Proc. NeurIPS*, vol. 32, 2019.
- [13] T. Yuan et al., "Token Labeling: Training with Label at Each Token," in *Proc. ECCV*, 2022.

- [14] H. Fang et al., “Eva: Exploring the Limits of Masked Visual Representation Learning at Scale,” *arXiv preprint arXiv:2211.07636*, 2022.
- [15] S. Han et al., “Transformer in Vision: A Survey,” *ACM Comput.Surv.*, vol. 55, no. 8, pp. 1–37, 2023.